

Quarter I review paper · Algebra

Grade 11

40 minutes 100 marks 14 questions + bonus No calculator

1–2	Increments, and the problem of the tangent	Algebra 11, §1.1 · P1 · 7.1
3–4	The limit of a function	Algebra 11, §1.2 · P1 · 7.1
3–4 ext	Limits at infinity — extension	Algebra 11, §1.2 · P1 · 7.1 · new on this paper
5–6	The derivative of a function	Algebra 11, §1.3 · P1 · 7.1–7.2
7–9	The rules of differentiation	Algebra 11, §1.4 · P1 · 7.2–7.4
10–12	The derivative of a composite function	Algebra 11, §1.5 · P1 · 7.5
15–16	The modulus function	P1 · 1.6 · P2 · 1.1–1.3
17–18	The equations of the tangent and the normal	Algebra 11, §1.6 · P1 · 7.6
19–22	Investigating a function with the derivative	Algebra 11, §1.7 · P1 · 7.7–7.8
23–25	Extremum problems	Algebra 11, §1.8 · P1 · 7.8

- Show your working — method carries marks. A limit or a derivative written down with no working scores nothing.
- For a limit at infinity, divide numerator and denominator by the highest power of x and say which terms you are sending to zero.
- For a stationary point, state its nature and the test you used.

What the paper weighs

TOPIC	MARKS	SHARE
Lessons 1–2 · Increments, and the problem of the tangent	5	5%
Lessons 3–4 · The limit of a function	5	5%
Lessons 3–4, extension · Limits at infinity	23	23%
Lessons 5–6 · The derivative of a function	8	8%
Lessons 7–9 · The rules of differentiation	13	13%
Lessons 10–12 · The derivative of a composite function	13	13%
Lessons 15–16 · The modulus function	5	5%
Lessons 17–18 · The equations of the tangent and the normal	8	8%
Lessons 19–22 · Investigating a function with the derivative	10	10%
Lessons 23–25 · Extremum problems	10	10%

Limits at infinity carry 23 % because the paper introduces them. The derivative block — the five rules, the chain rule, the tangent, the investigation and the extremum — carries 54 % together, which is the share of the quarter it occupied. The modulus is examined once, at five marks: it is a Cambridge insert, not part of the derivative thread.

A Section A · Five marks each

One or two lines of working is enough.

EASY

30 marks

- 1 Differentiate: $y = 4x^3 - 5x + 7$ [5]

ANSWER

$$\frac{dy}{dx} = 12x^2 - 5$$

WORKING

Power rule term by term; the derivative of the constant 7 is zero.

- 2 Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$ [5]

ANSWER

3

WORKING

Divide top and bottom by x^2 : $\frac{3 + 2/x}{1 - 5/x^2}$. Both $\frac{2}{x}$ and $\frac{5}{x^2}$ tend to 0, leaving $\frac{3}{1} = 3$.

Equal degrees, so the answer is the ratio of the leading coefficients — but the division has to be shown, not quoted.

- 3 Evaluate: $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ [5]

ANSWER

8

WORKING

Substitution gives $\frac{0}{0}$. Factorise: $\frac{(x-4)(x+4)}{x-4} = x+4$ for $x \neq 4$, which tends to 8.

- 4 For $y = x^2$, the value of x changes from 3 to 3.1. Find Δy , and the gradient of the secant through the two points. [5]

ANSWER

$$\Delta y = 0.61; \text{ gradient} = 6.1$$

WORKING

$$\Delta y = 3.1^2 - 3^2 = 9.61 - 9 = 0.61 \text{ and } \Delta x = 0.1, \text{ so } \frac{\Delta y}{\Delta x} = \frac{0.61}{0.1} = 6.1.$$

The tangent gradient at $x = 3$ is 6; the secant is close but not equal, which is the whole point of lessons 1–2.

- 5 Differentiate: $y = (3x + 1)^5$ [5]

ANSWER

$$\frac{dy}{dx} = 15(3x + 1)^4$$

WORKING

Chain rule: $5(3x + 1)^4$ times the derivative of the inside, which is 3.

6 Solve: $|2x - 6| = 10$ [5]

ANSWER

$x = 8$ or $x = -2$

WORKING

$2x - 6 = 10$ gives $x = 8$; $2x - 6 = -10$ gives $x = -2$. Both satisfy the original equation.

One answer only means the negative branch was forgotten.

B Section B · Eight marks each

MEDIUM

40 marks

Method marks are available even when the final answer is wrong.

7 Evaluate, showing the division in each case:

(a) $\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 1}{5x^2 + 4}$ [3]

(b) $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 - 2}$ [2]

(c) Explain why $\lim_{x \rightarrow \infty} \frac{x^3 + 2x}{4x^2 - 1}$ does not exist. [3]

ANSWER

(a) $\frac{2}{5}$ (b) 0 (c) it increases without bound

WORKING

(a) Divide by x^2 : $\frac{2 - 3/x + 1/x^2}{5 + 4/x^2} \rightarrow \frac{2}{5}$.

(b) Divide by x^3 : $\frac{1/x + 1/x^3}{1 - 2/x^3} \rightarrow \frac{0}{1} = 0$.

(c) Divide by x^2 : $\frac{x + 2/x}{4 - 1/x^2}$. The denominator tends to 4, but the numerator contains x , which grows without bound — so the quotient grows without bound and no finite limit exists.

The three parts are the three degree cases: top = bottom, top < bottom, top > bottom. A student who has only learnt “compare the degrees” can state the answers but cannot earn the method marks.

8 $f(x) = x^2 - 3x$.

(a) Show that $\frac{f(2+h) - f(2)}{h} = h + 1$. [5]

(b) Hence write down $f'(2)$. [3]

ANSWER

(a) shown below (b) $f'(2) = 1$

WORKING

(a) $f(2+h) = (2+h)^2 - 3(2+h) = 4 + 4h + h^2 - 6 - 3h = h^2 + h - 2$, and $f(2) = 4 - 6 = -2$. So the numerator is $h^2 + h$, and dividing by h gives $h + 1$ — legal because $h \neq 0$ throughout.

(b) As $h \rightarrow 0$, $h + 1 \rightarrow 1$, so $f'(2) = 1$.

Cancelling the h before noting $h \neq 0$ is the slip to look for. Checking with the rules, $f'(x) = 2x - 3$ and $f'(2) = 1$, is worth a mark but is not the method asked for.

- 9 Differentiate $y = \frac{2x-1}{x^2+3}$ and find the gradient of the curve at $x = 1$. [8]

ANSWER

$$\frac{dy}{dx} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}; \text{ gradient} = \frac{3}{8}$$

WORKING

Quotient rule with $u = 2x - 1$, $v = x^2 + 3$, so $u' = 2$ and $v' = 2x$:

$$\frac{dy}{dx} = \frac{2(x^2 + 3) - (2x - 1)(2x)}{(x^2 + 3)^2} = \frac{2x^2 + 6 - 4x^2 + 2x}{(x^2 + 3)^2} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}.$$

$$\text{At } x = 1: \frac{-2 + 2 + 6}{4^2} = \frac{6}{16} = \frac{3}{8}.$$

The expansion $-(2x - 1)(2x) = -4x^2 + 2x$ is where the sign is usually lost.

- 10 The curve $y = x^3 - 4x$. Find the equation of the tangent and the equation of the normal at the point where $x = 2$. [8]

ANSWER

$$\text{tangent } y = 8x - 16; \text{ normal } x + 8y = 2$$

WORKING

At $x = 2$, $y = 8 - 8 = 0$, so the point is $(2, 0)$. $\frac{dy}{dx} = 3x^2 - 4$, which is 8 at $x = 2$.

Tangent: $y - 0 = 8(x - 2)$, so $y = 8x - 16$.

Normal gradient $= -\frac{1}{8}$: $y - 0 = -\frac{1}{8}(x - 2)$, so $8y = -x + 2$, that is $x + 8y = 2$.

Using -8 rather than $-\frac{1}{8}$ for the normal is the standard error; the product of the two gradients must be -1 .

- 11 $y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ and its value at $x = 4$. [8]

ANSWER

$$\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 9}}; \text{ at } x = 4 \text{ it is } \frac{4}{5}$$

WORKING

Write $y = (x^2 + 9)^{1/2}$. Chain rule: $\frac{dy}{dx} = \frac{1}{2}(x^2 + 9)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + 9}}$.

At $x = 4$: $\sqrt{16 + 9} = 5$, so the value is $\frac{4}{5}$.

The factor $2x$ from the inside function is the mark most often dropped.

C Section C • Ten marks each

Full method required. Define your variable before you differentiate.

HARD

30 marks

12 $y = x^3 - 6x^2 + 9x$.

(a) Find the stationary points and determine the nature of each. [6]

(b) State the intervals on which the function is increasing and decreasing, and sketch the curve.

[4]

ANSWER

(a) (1, 4) maximum, (3, 0) minimum (b) increasing for $x < 1$ and $x > 3$, decreasing for $1 < x < 3$

WORKING

(a) $\frac{dy}{dx} = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, zero at $x = 1$ and $x = 3$. Then $y(1) = 1 - 6 + 9 = 4$ and

$y(3) = 27 - 54 + 27 = 0$.

Second derivative $\frac{d^2y}{dx^2} = 6x - 12$: at $x = 1$ it is $-6 < 0$, a maximum; at $x = 3$ it is $6 > 0$, a minimum.

(b) $3(x-1)(x-3) > 0$ for $x < 1$ and for $x > 3$, so the function increases there, and < 0 for $1 < x < 3$, so it decreases between the two stationary points. The curve also passes through the origin, since $y = x(x-3)^2$, and rises without bound on the right.

A sign chart earns the nature marks just as well as the second derivative. The sketch must show the maximum to the *left* of the minimum — the commonest sketch error is drawing them the other way round.

- 13 A closed cylinder is to hold $128\pi \text{ cm}^3$. Find the radius that makes its total surface area least, and state that least area in terms of π . Show that your value gives a minimum. [10]

ANSWER

$r = 4 \text{ cm}$, $h = 8 \text{ cm}$, least area $96\pi \text{ cm}^2$

WORKING

Volume: $\pi r^2 h = 128\pi$, so $h = \frac{128}{r^2}$.

Surface area: $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \cdot \frac{128}{r^2} = 2\pi r^2 + \frac{256\pi}{r}$.

$\frac{dS}{dr} = 4\pi r - \frac{256\pi}{r^2} = 0$ gives $4\pi r^3 = 256\pi$, so $r^3 = 64$ and $r = 4$.

Then $h = \frac{128}{16} = 8$ and $S = 2\pi(16) + \frac{256\pi}{4} = 32\pi + 64\pi = 96\pi$.

$\frac{d^2S}{dr^2} = 4\pi + \frac{512\pi}{r^3}$, which is $12\pi > 0$ at $r = 4$, so the area is a minimum.

The marks are in the modelling: eliminating h with the volume before differentiating. A student who differentiates S with both r and h in it cannot finish.

14

- (a) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 1}}{2x + 5}$. [4]
- (b) Evaluate $\lim_{x \rightarrow \infty} (\sqrt{4x^2 + 3x} - 2x)$. [6]

ANSWER

(a) $\frac{3}{2}$ (b) $\frac{3}{4}$

WORKING

(a) Divide by x , taking $x > 0$ so that $x = \sqrt{x^2}$:

$$\frac{\sqrt{9x^2 + 1}}{2x + 5} = \frac{\sqrt{9 + 1/x^2}}{2 + 5/x} \rightarrow \frac{\sqrt{9}}{2} = \frac{3}{2}.$$

(b) The difference is an $\infty - \infty$ form, so multiply by the conjugate:

$$\sqrt{4x^2 + 3x} - 2x = \frac{(4x^2 + 3x) - 4x^2}{\sqrt{4x^2 + 3x} + 2x} = \frac{3x}{\sqrt{4x^2 + 3x} + 2x}.$$

$$\text{Now divide by } x: \frac{3}{\sqrt{4 + 3/x} + 2} \rightarrow \frac{3}{2 + 2} = \frac{3}{4}.$$

Part (b) is the conjugate trick from lesson 3–4 used at infinity instead of at a point. Answering 0 means the subtraction was done term by term, which $\infty - \infty$ does not allow.

Bonus



Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100.

BONUS

+10 marks

15

$$y = \frac{8x}{x^2 + 4} \text{ for } x \geq 0.$$

- (a) Show that the curve has a maximum at $x = 2$ and find its value. [6]
- (b) Find $\lim_{x \rightarrow \infty} y$ and say what it tells you about the shape of the curve. [4]

ANSWER

(a) maximum (2, 2) (b) 0; the curve falls back towards the x -axis, which is a horizontal asymptote

WORKING

(a) Quotient rule: $\frac{dy}{dx} = \frac{8(x^2 + 4) - 8x(2x)}{(x^2 + 4)^2} = \frac{32 - 8x^2}{(x^2 + 4)^2} = \frac{-8(x - 2)(x + 2)}{(x^2 + 4)^2}.$

The denominator is always positive, so the sign is that of $-8(x - 2)(x + 2)$: positive for $0 \leq x < 2$ and negative for $x > 2$. The function rises then falls, so $x = 2$ is a maximum, and $y(2) = \frac{16}{8} = 2$.

(b) Divide by x^2 : $\frac{8/x}{1 + 4/x^2} \rightarrow \frac{0}{1} = 0$. So after the maximum the curve decreases towards the x -axis without ever reaching it — $y = 0$ is a horizontal asymptote.

This is the quarter in one question: a quotient-rule derivative, a stationary point with its nature, and a limit at infinity read as the shape of the curve. The sign argument is cleaner here than the second derivative, which is why it is the working given — but $\frac{d^2y}{dx^2} = -\frac{1}{2}$ at $x = 2$ earns the same marks.